

0B.M.S. COLLEGE OF ENGINEERING BENGALURU
Autonomous Institute, Affiliated to VTU



Lab Record

Object Oriented Analysis and Design

Submitted in partial fulfillment for the 5th Semester Laboratory

Bachelor of Technology
in
Computer Science and Engineering

Submitted by:

Dhanush K

(1BM24CS404)

Department of Computer Science and Engineering
B.M.S. College of Engineering
Bull Temple Road, Basavanagudi, Bangalore 560 019
Aug-Dec 2025

B.M.S. COLLEGE OF ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE AND
ENGINEERING



CERTIFICATE

This is to certify that the Object-Oriented Analysis and Design(23CS6PCSEO) laboratory has been carried out by Dhanush K (1BM24CS404) during the 5th Semester Aug-Dec-2025.

Signature of the Faculty Incharge:

NAME OF THE FACULTY: RoopaShree CS

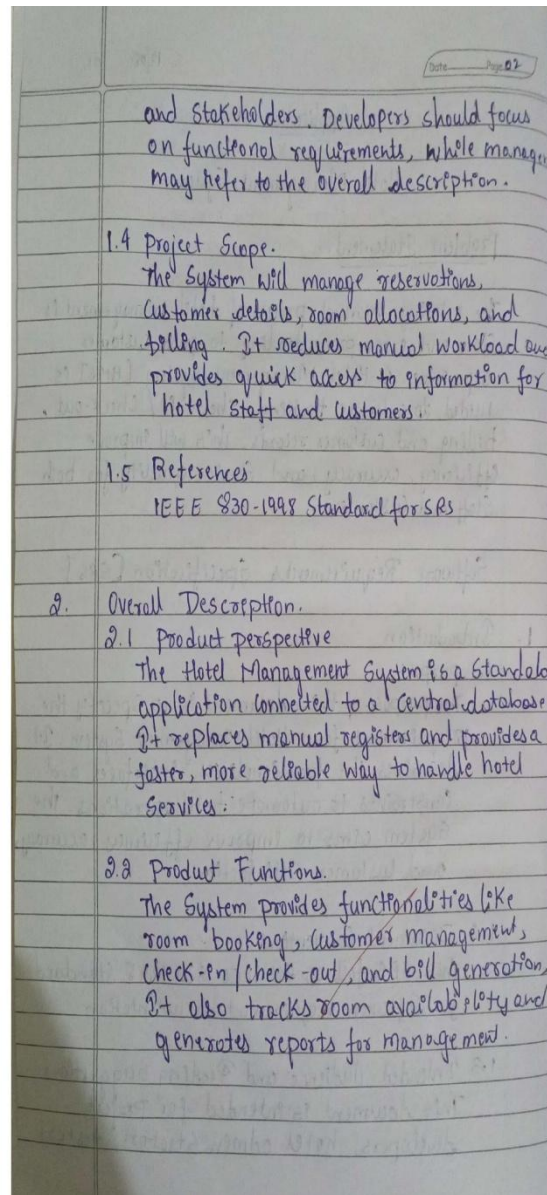
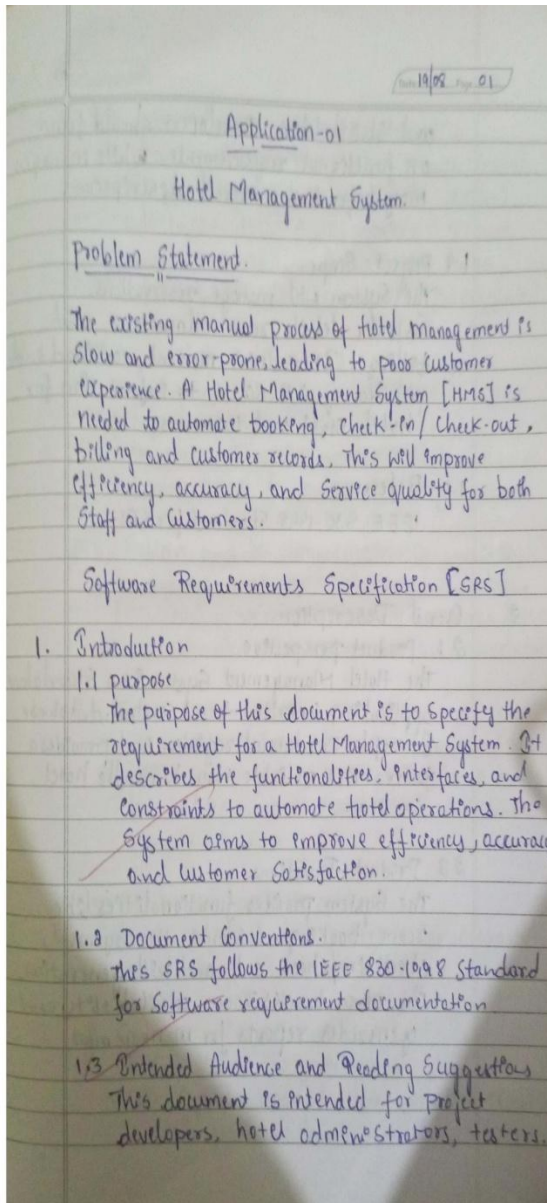
Department of Computer Science and Engineering
B.M.S. College of Engineering, Bangalore

Table of Contents

Sl no	content	Pg no
1.	Hotel Management System	4-8
2.	Credit Card Processing System	9-13
3.	Library Management System	14-18
4.	Stock Maintenance System	19-23
5.	Passport Automation System	24-28

1. Hotel Management System

1.1 SRS DOCUMENT



2.3 User classes and characteristics
There are three main user groups: hotel staff (who operate the system daily), administrators (who manage records and reports), and customers (who may use a limited booking interface).

2.4 Operating Environment
The system will run on Windows or Linux platforms with an SQL database. It requires a basic computer setup and internet access for online booking support.

2.5 Design and Implementation Constraints.
The system must use a relational database for storing records. Security measures like authentication and encryption are mandatory.

2.6 User Documentation.
The system will include a user manual and quick-start guide. Training sessions may also be provided to hotel staff to ensure smooth adoption.

2.7 Assumptions and Dependencies.
It is assumed that internet services and power supply will be available.

3. Specific Requirements.

3.1 Functional Requirements.

- The system should allow room reservation and cancellation.

- check-in / check-out → customers must be able to check-in / check-out through the system.
- automatic bill → It should support with tax calculation.
- room availability status → The system must show in real time.

3.2 External Interface Requirements.

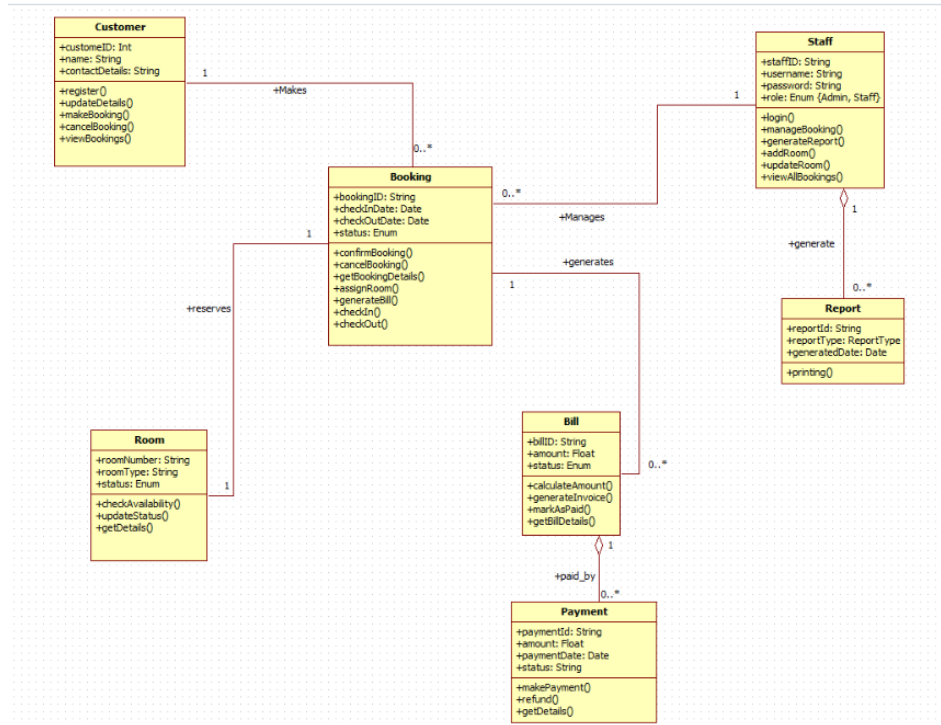
- User Interfaces → login screen, booking forms, and billing screens should be simple and interactive.
- Hardware Interfaces → The system should support desktops and printers for invoice printing.
- Software Interface → Must integrate with SQL databases.

3.3 Non-functional Requirements.

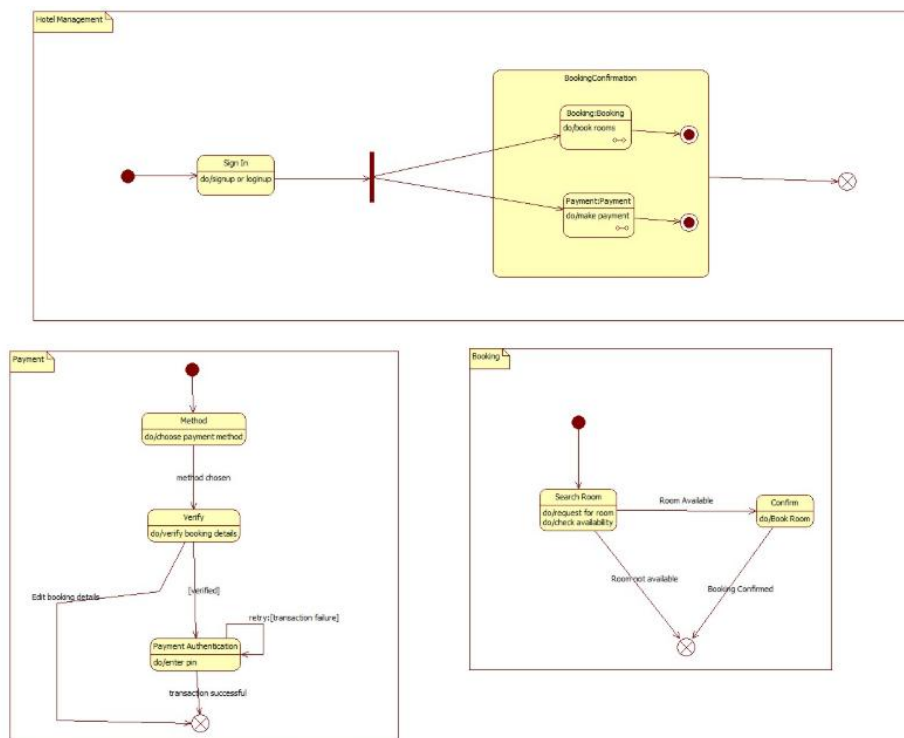
- Reliability → The system must perform consistently without crashes.
- Availability → It should be available 24/7 for customers & staff.
- Security → Data must be encrypted and user login should be password-protected.
- Performance → Response time should be less than 2 seconds per transaction.

4. Appendices.

1.2 CLASS DIAGRAM

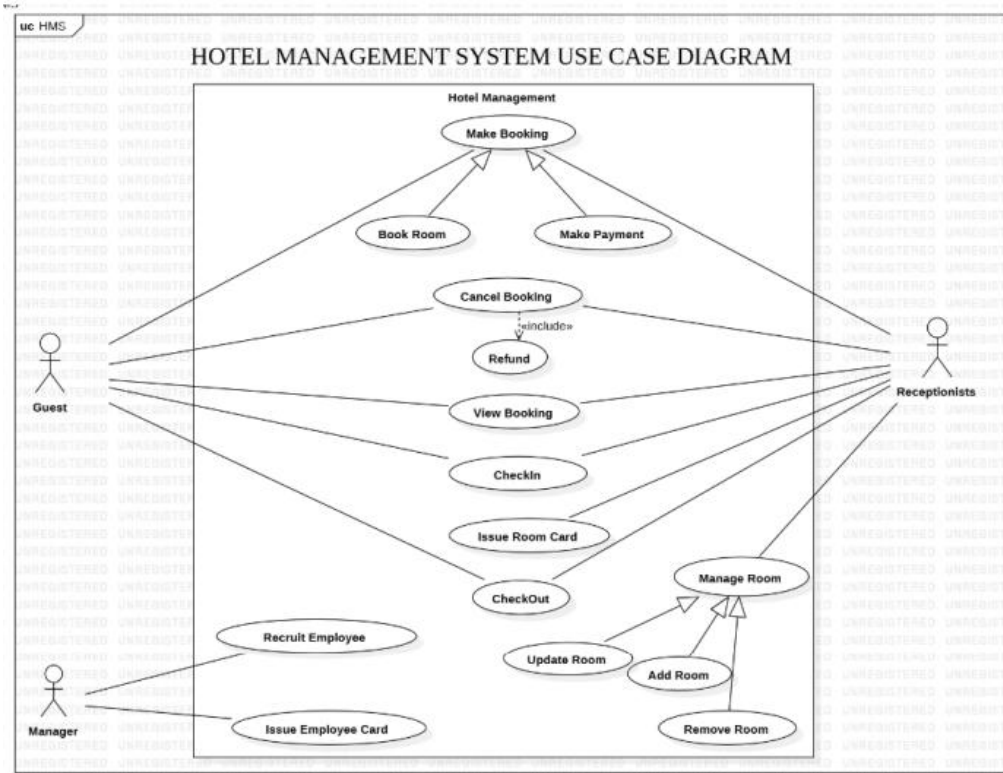


1.3 STATE DIAGRAM

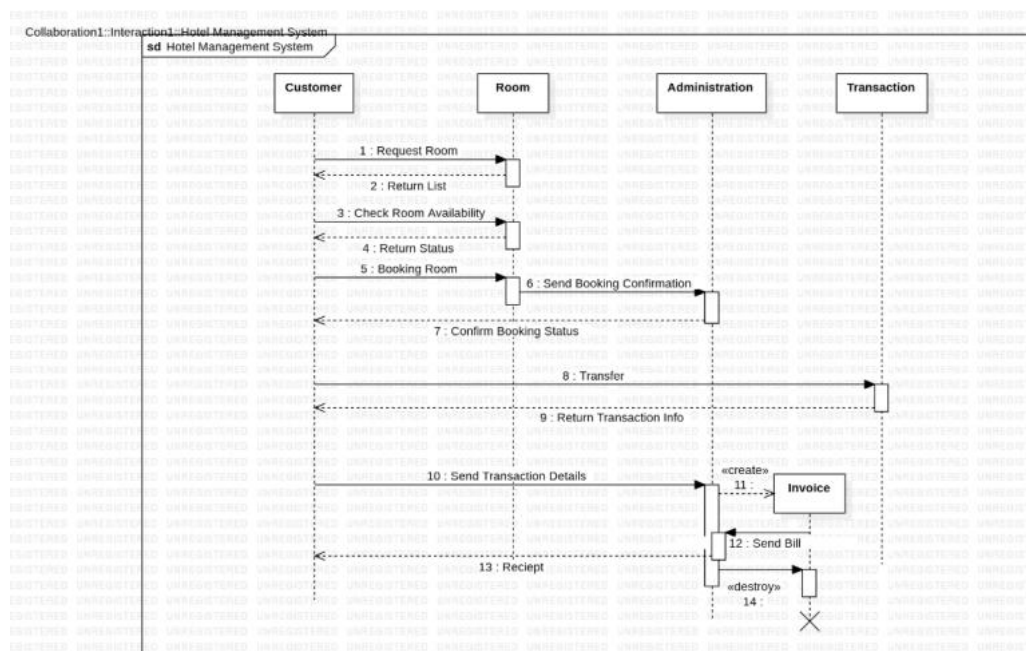


1.4 INTERACTION MODELS

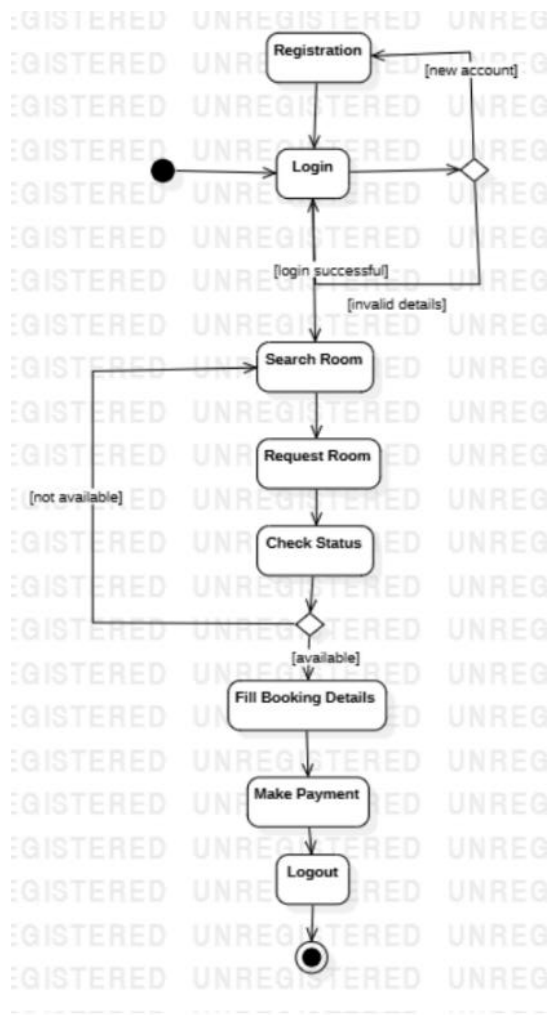
a. Use case model



b. Sequence model

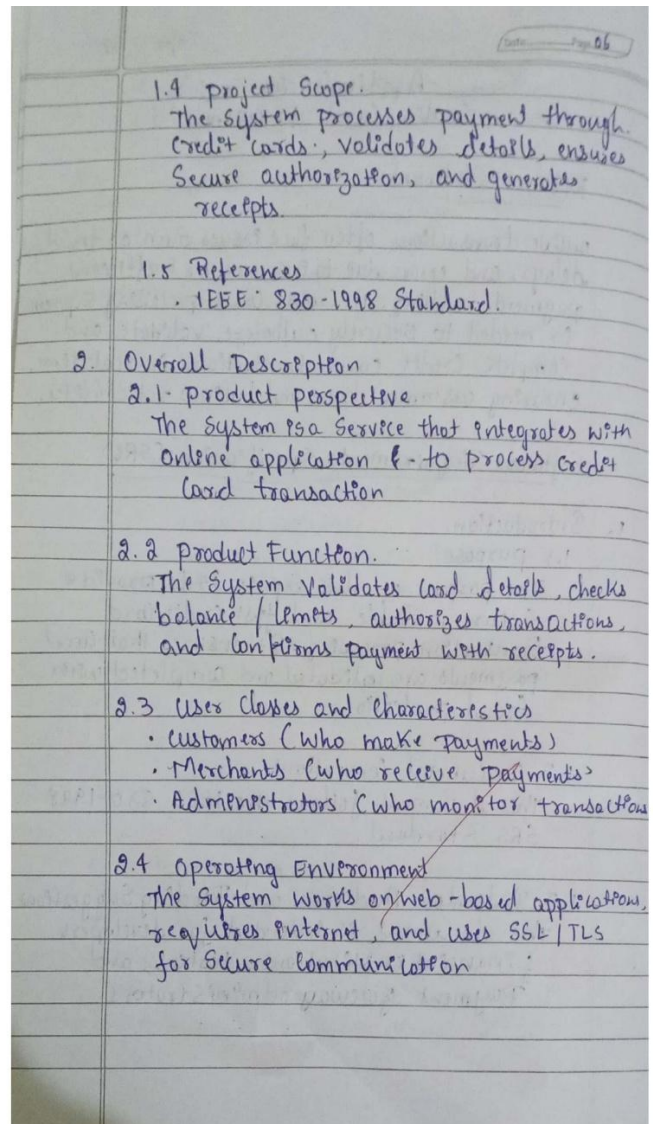
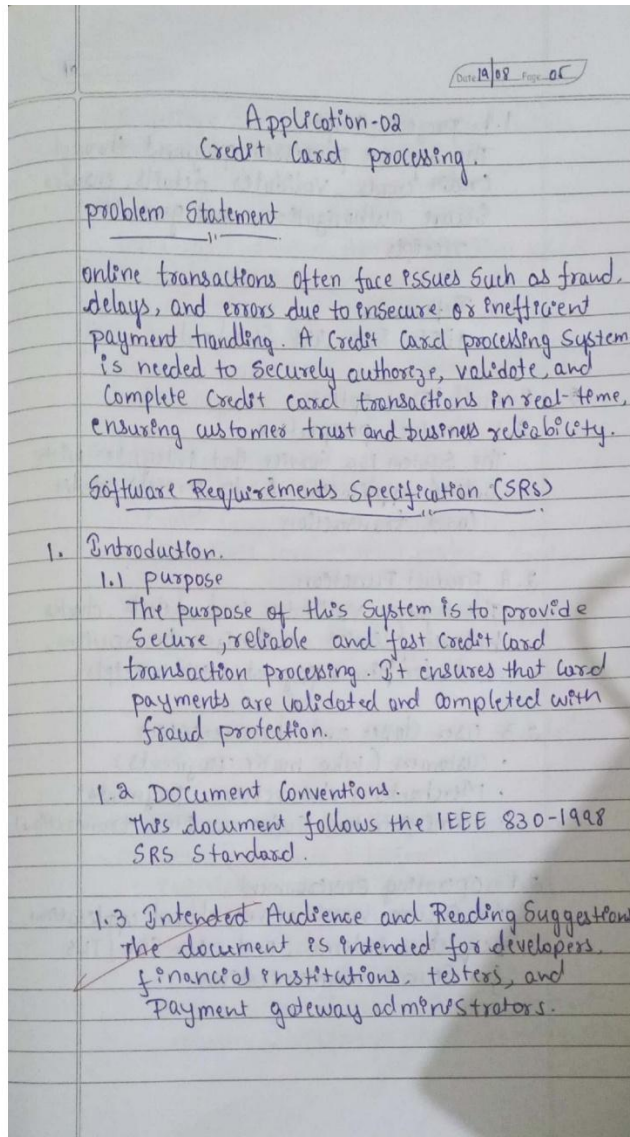


c. Activity model



2. Credit Card Processing

2.1 SRS DOCUMENT



2.5 Design and Implementation Constraints.
Must follow PCI DSS Compliance.

2.6 User Documentation
User manuals and API integration guides will be provided.

2.7 Assumptions and Dependencies.
Assumes stable internet connection and active bank / payment gateway services.

3. Specific Requirements

3.1 Key Functional Requirements.

- Credit Card payment authorization
- Real-time Card → must perform card validation
- Transaction logging → Should support for auditing.
- Must generate digital receipts → for every transaction
- Refunds and cancellation → Should support

3.2 External Interface Requirements.

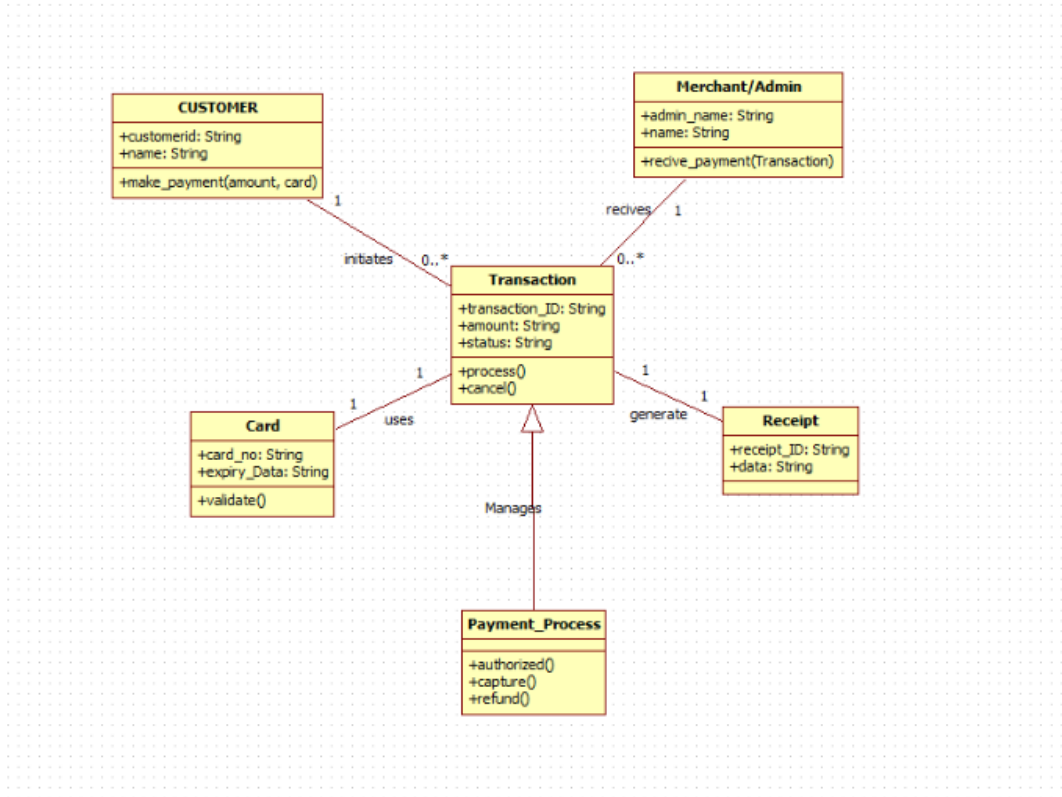
- User Interfaces → Payment form for entering card details, error messages, and receipt screens.
- Hardware Interfaces → optional POS terminal and receipt printers
- Software Interfaces → Integration with payment gateways and banking APIs.

3.3 Non-Functional Requirement

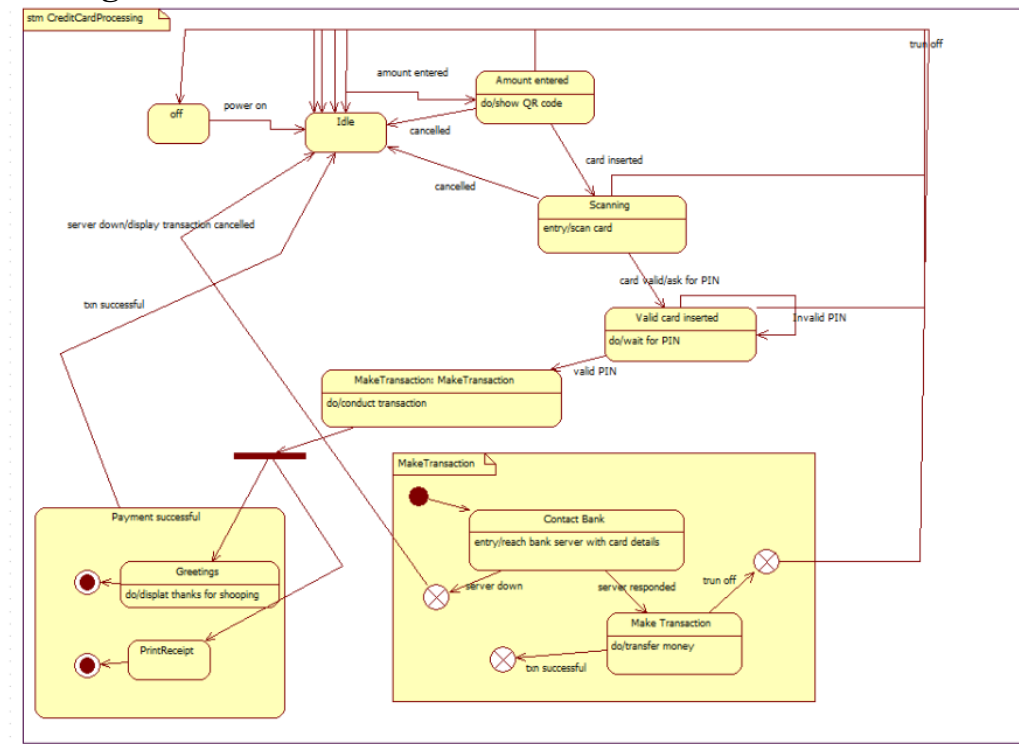
- Security → Must use encryption for card details
- Reliability → Transaction should not be lost or duplicated.
- Performance → Each transaction must complete within 5 seconds

4. Appendices.

2.3 Class Diagram

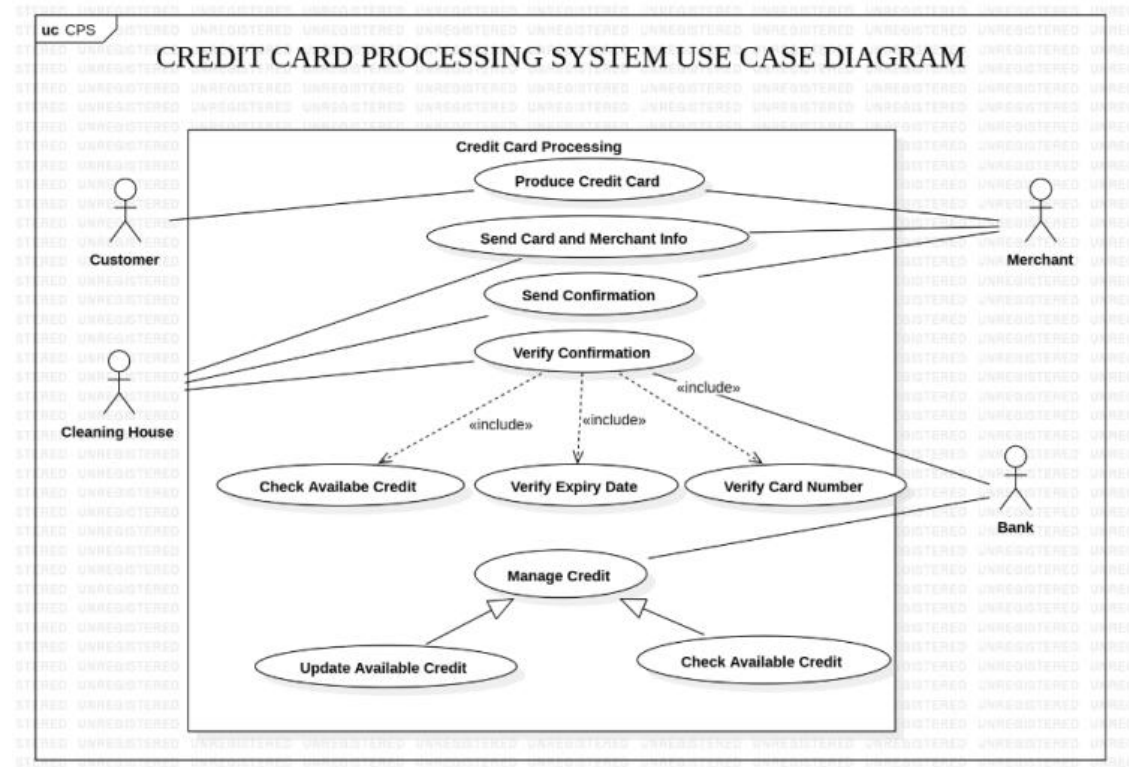


2.4 State Diagram

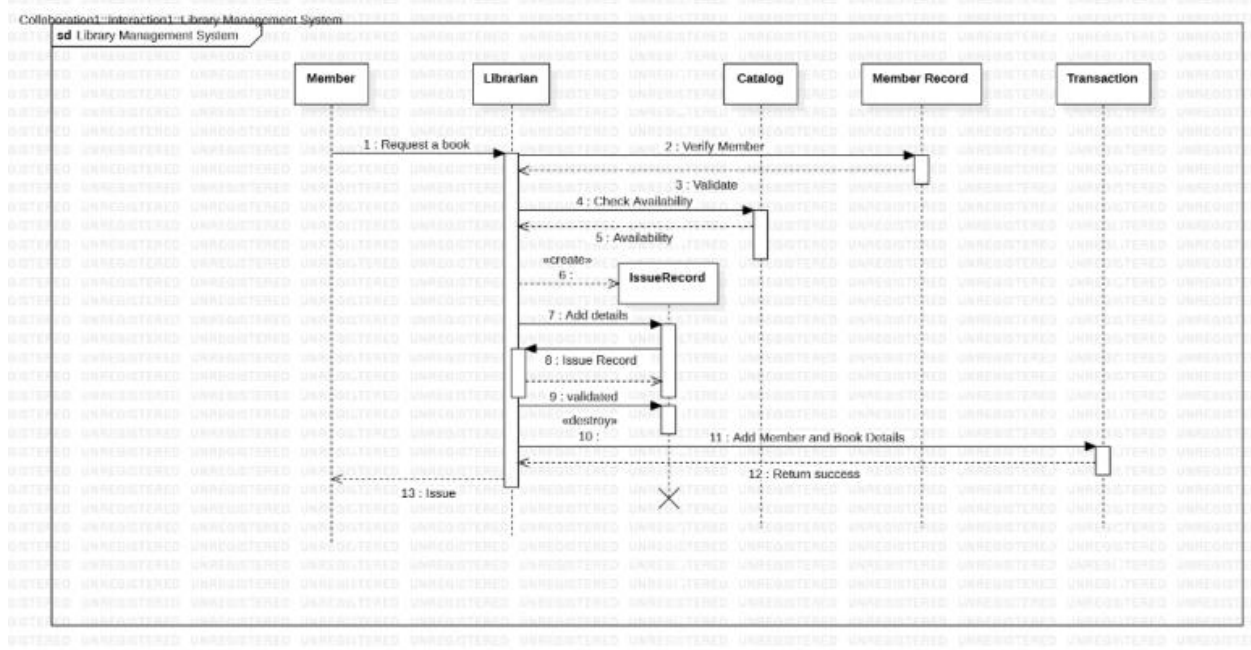


2.5 Interaction Models

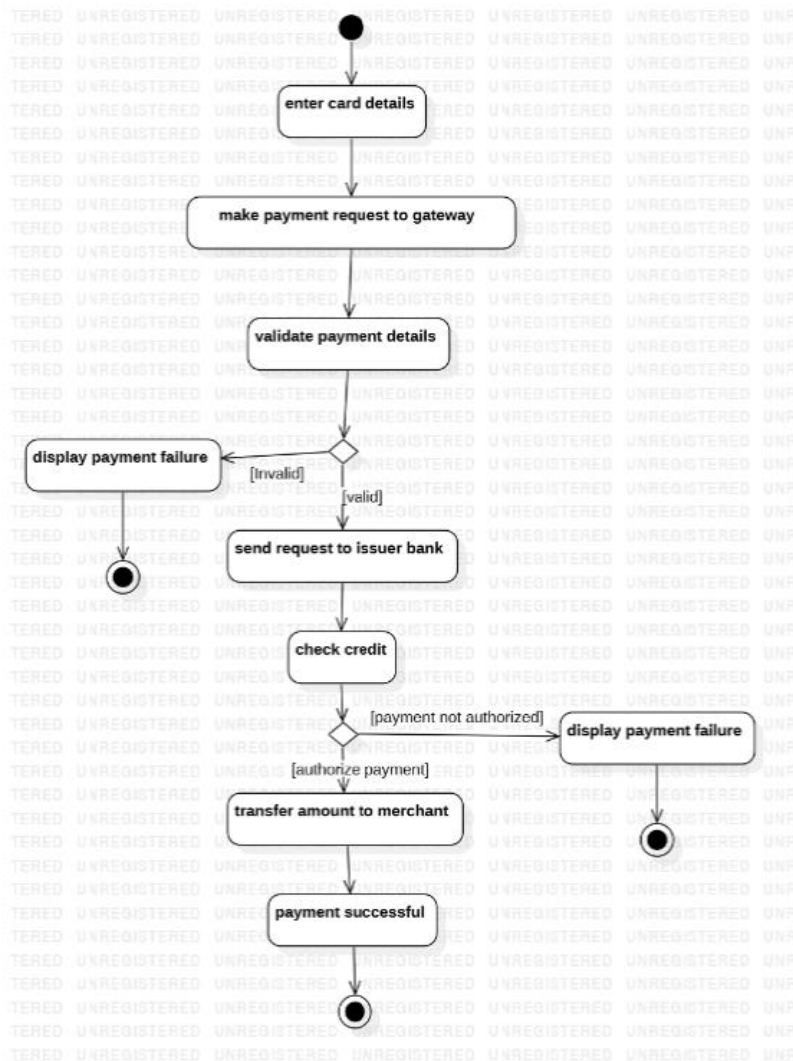
a. Use Case Model



d. Sequence model

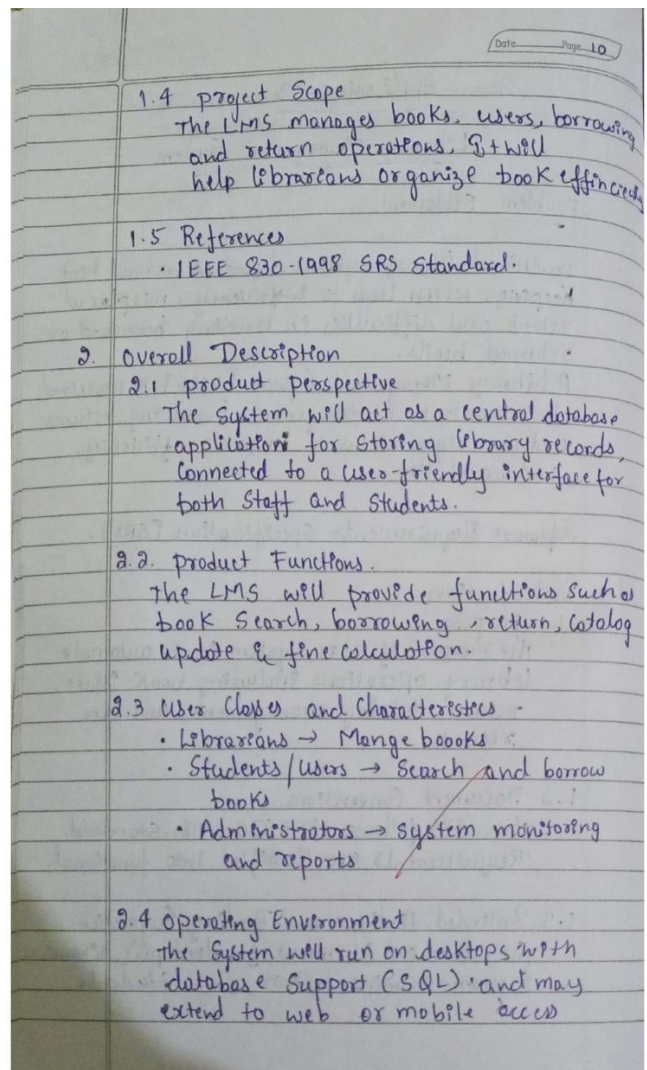
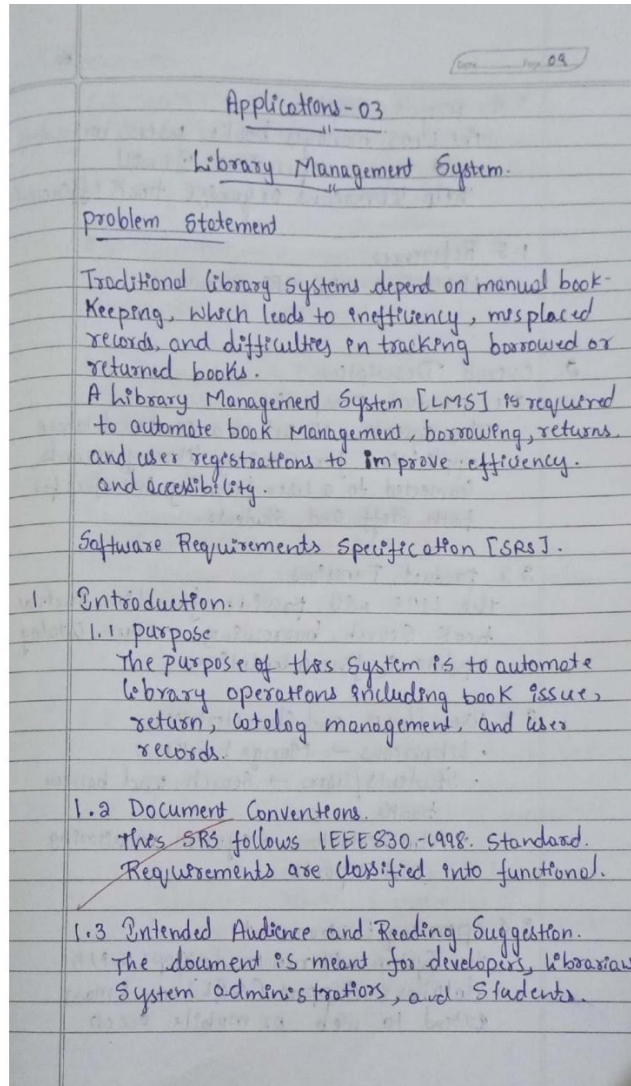


c. Activity Model



3. Library management system

3.1 SRS Document



2.5 Design and Implementation constraints
The System must use a relational database.
Authentication and role-based access are required.

2.6 User Documentation.
User manuals and quick guides will be provided for staff and students.

2.7 Assumptions & Dependencies
It is assumed that a stable internet or local network connection will be available.

3. Specific Requirements

3.1 Functional Requirements.

- Add, update, and delete book records
- Borrow and return books with due dates.
- Calculate and apply fines for late returns
- Generate library usage reports

3.2 External Interface Requirements.

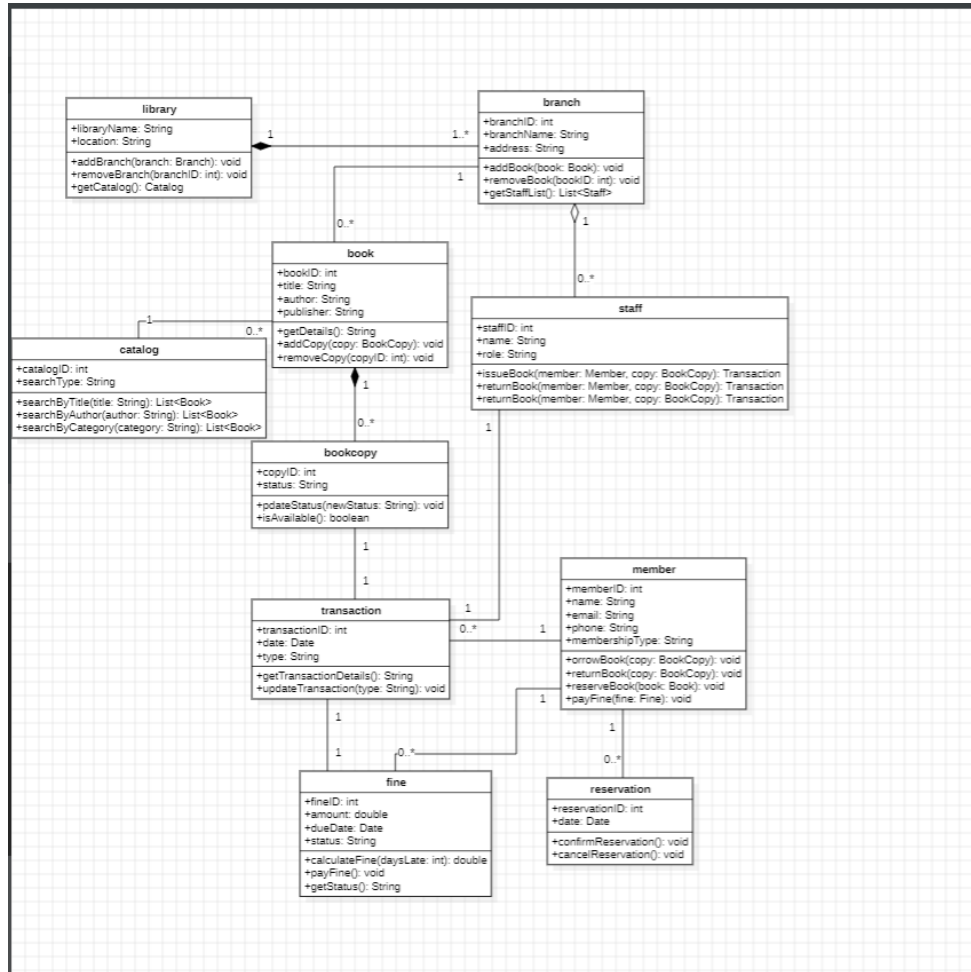
- User Interfaces → login screen, book search form, borrowing / return panel.
- Hardware Interfaces → Desktop computers, barcode scanner
- Software Interface → SQL Database, Student record system.
- Communication Interface: LAN / Internet for remote access.

3.3 Non-Functional Requirements

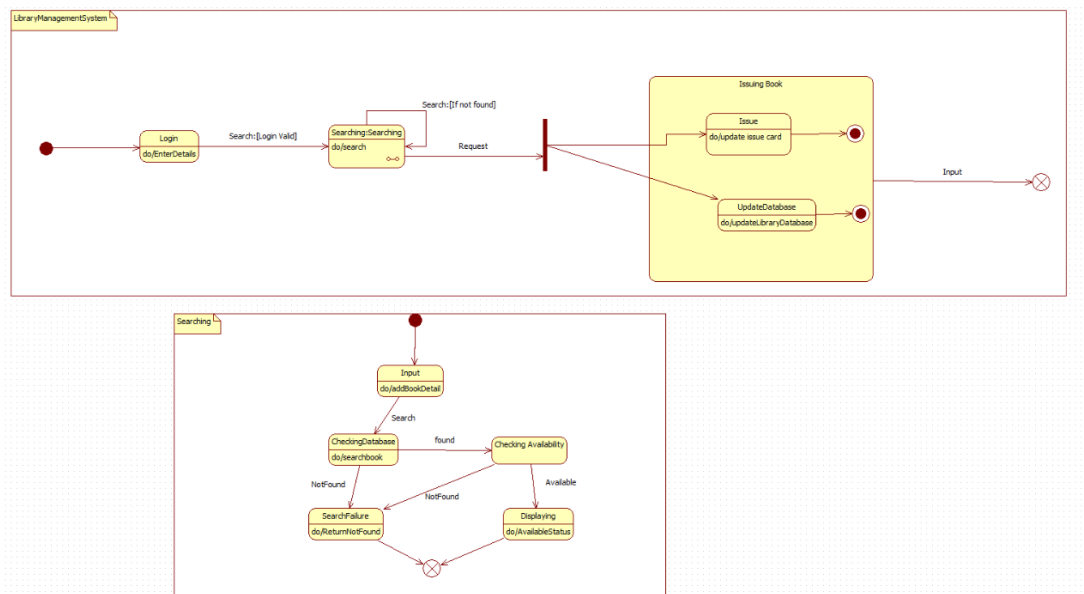
- User Interfaces → login screen, book search form, b
- Reliability → System must not lose records during update
- Availability → Accessible during library hours (24/7 for online portal)
- Security → User authentication and role-based access control.

4 Appendices

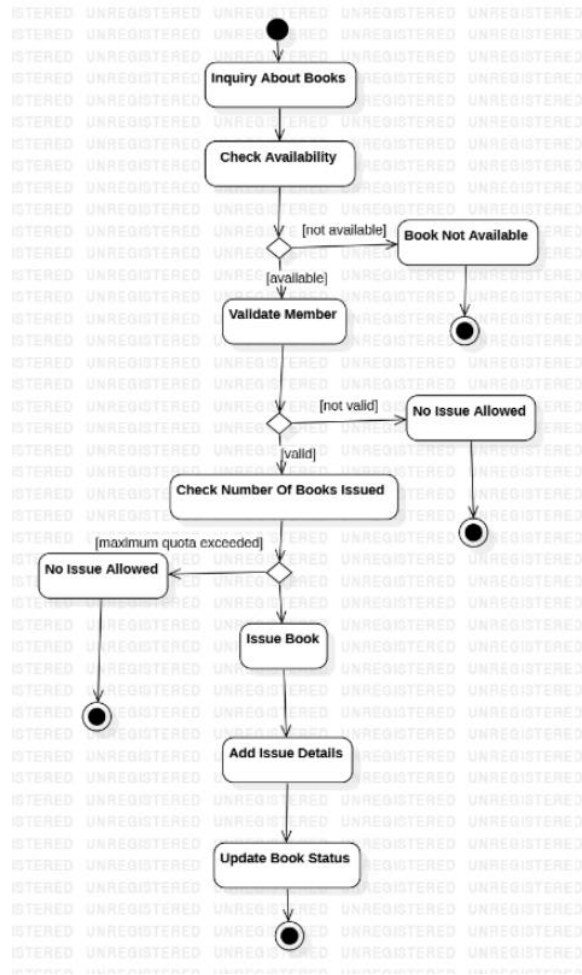
3.3 Class Model



3.4 State model

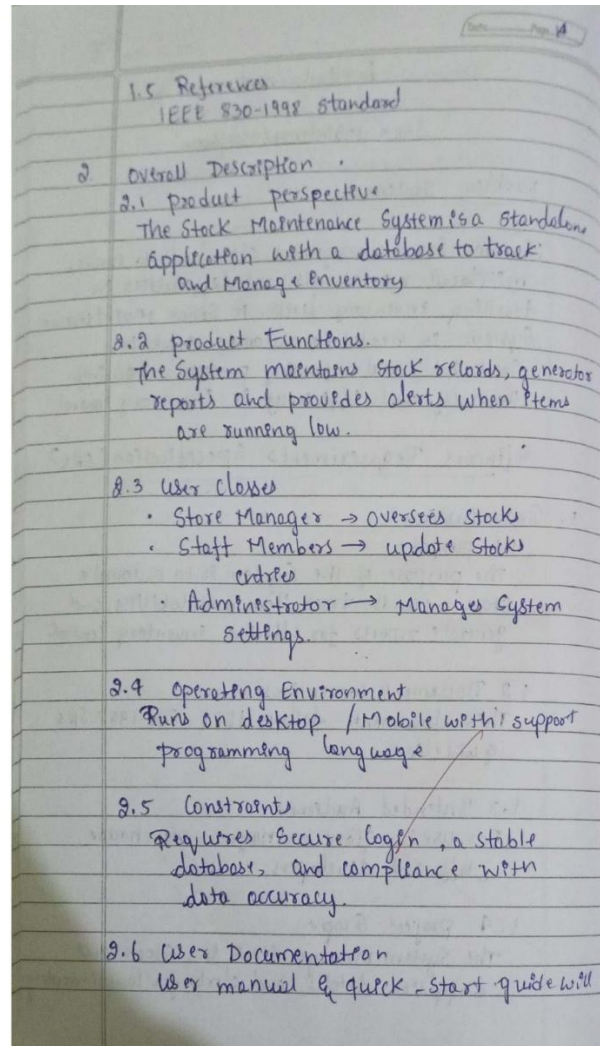
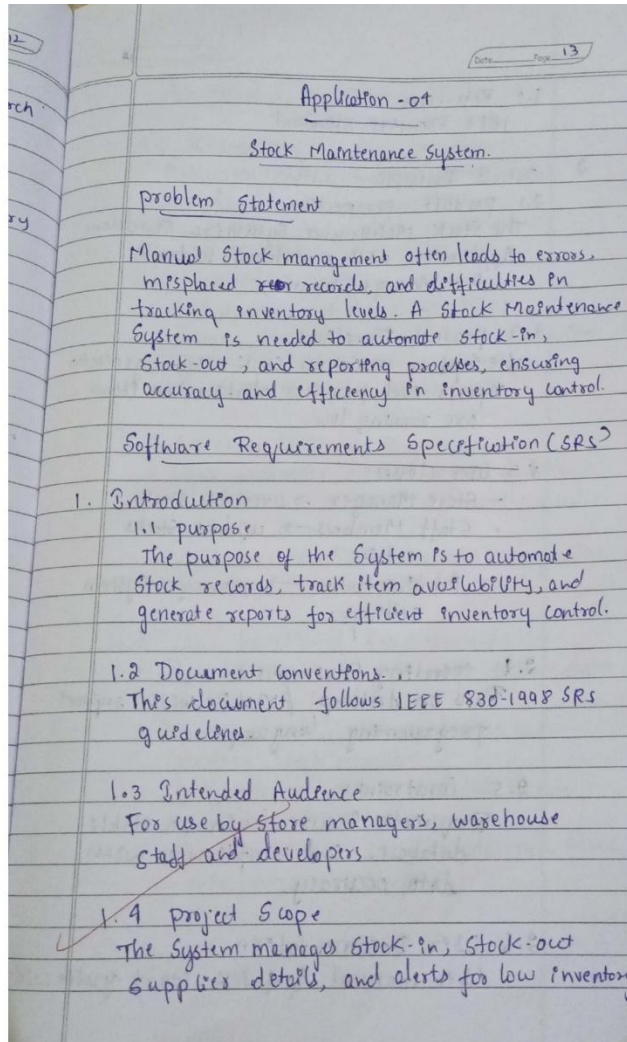


c. Activity Model



4. Stock Maintenance System

4.1 SRS Document



be provided

3.1 Assumptions & Dependencies
Assumes continuous internet connection
and functional hardware

3. Specific Requirements.

3.1 Functional Requirements.

- Add, update, and delete stock items
- Record stock-in and stock-out transaction.
- generate daily / weekly / monthly
- Send alerts for low stock levels.

3.2 External Interface Requirements

- User Interface → Dashboard for stock
- Hardware → Desktop / mobile device, barcode scanner
- Software → SQL database integration
- Communication → LAN / Internet

3.3 Non-functional Requirements

- Reliability → System must ensure data accuracy & consistency.
- Security → only authorized users can modify stock records
- Performance → stock updates should process within 2 seconds.
- Maintainability → easy to add new items, categories, and reports.

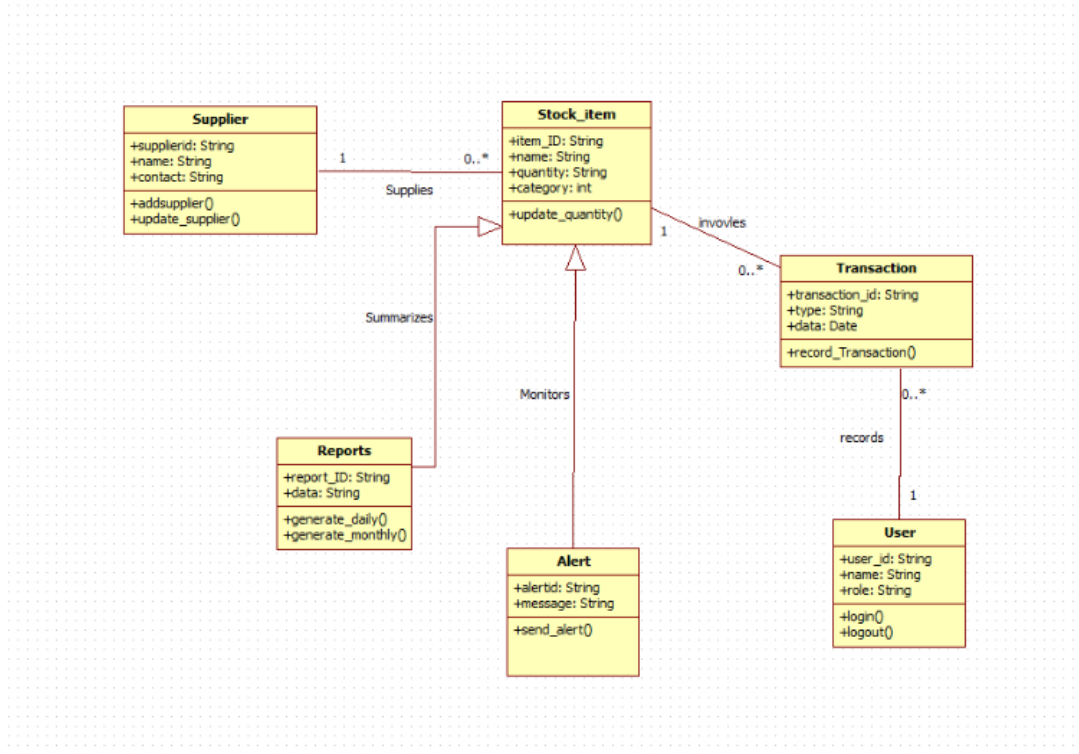
4 Appendices

→ SQL

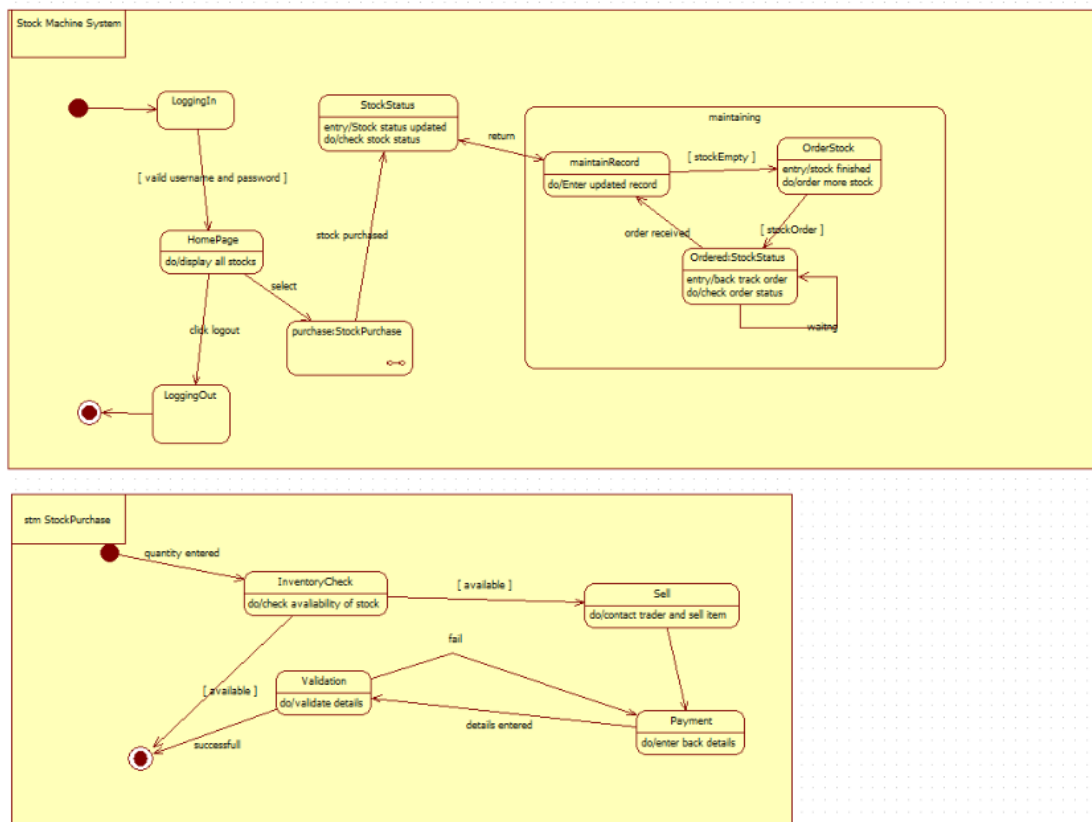
→ Stock in → Adding

→ Stock out → Removing.

4.3 Class model

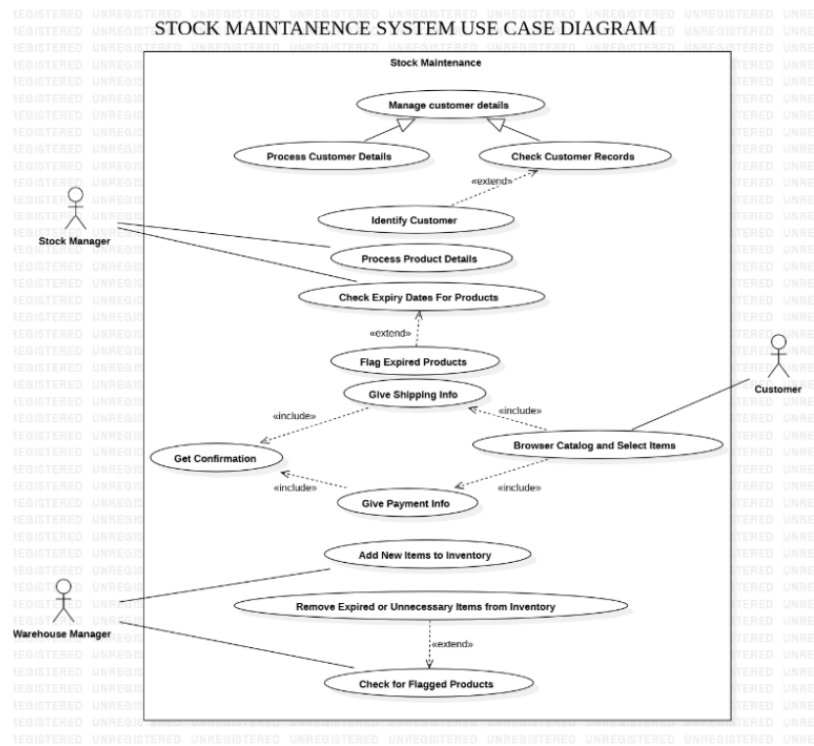


4.4 State Model

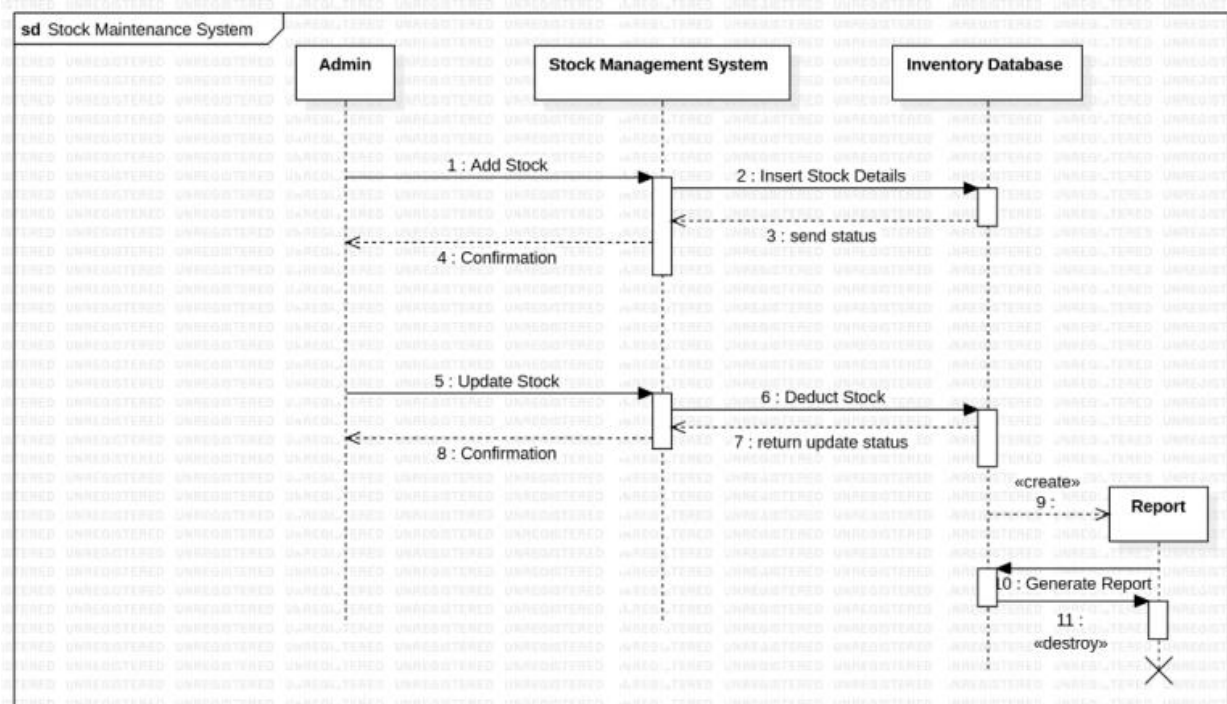


4.5 Interaction Models

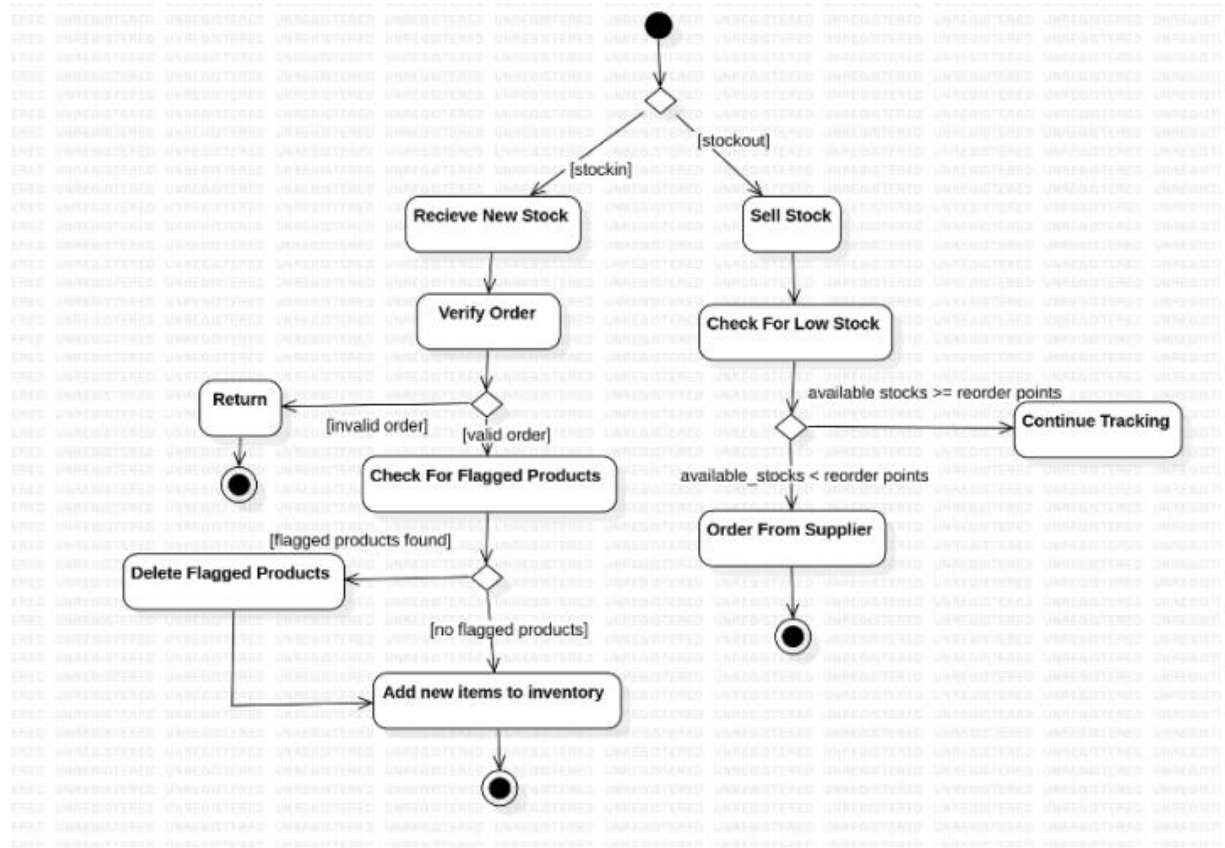
a. Use Case Model



d. Sequence Model

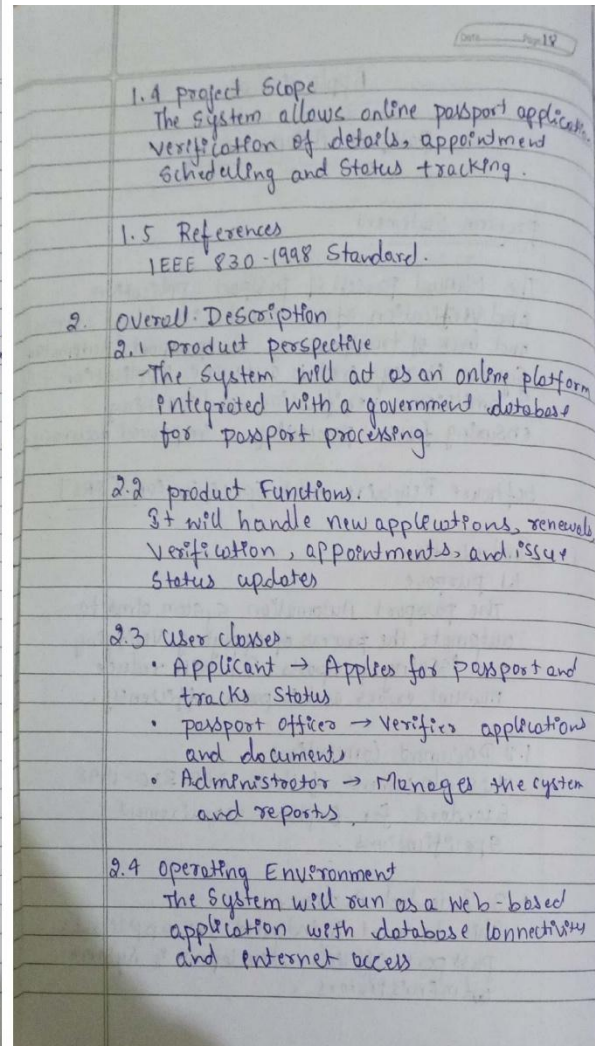
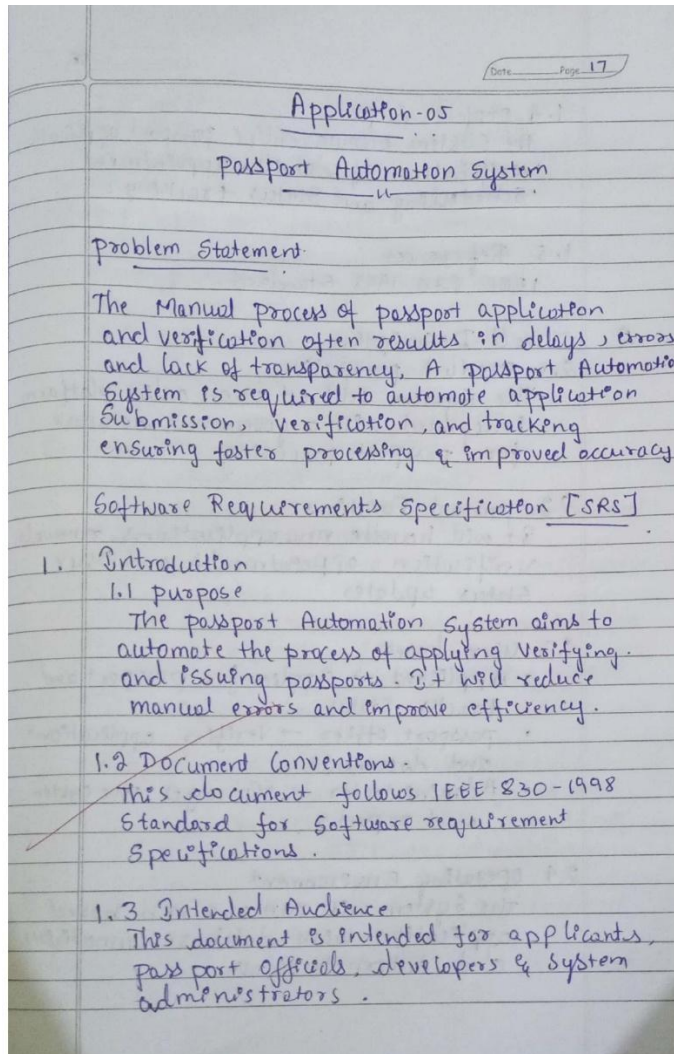


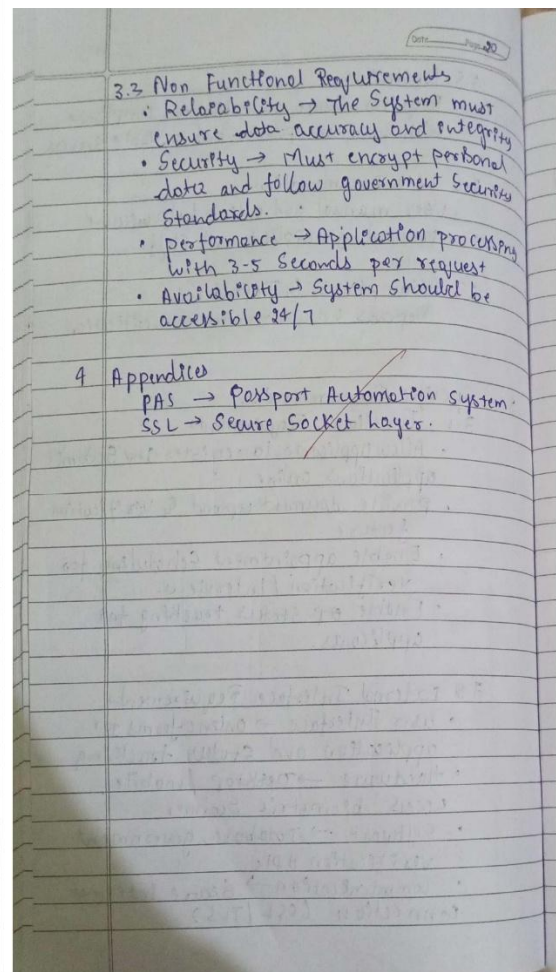
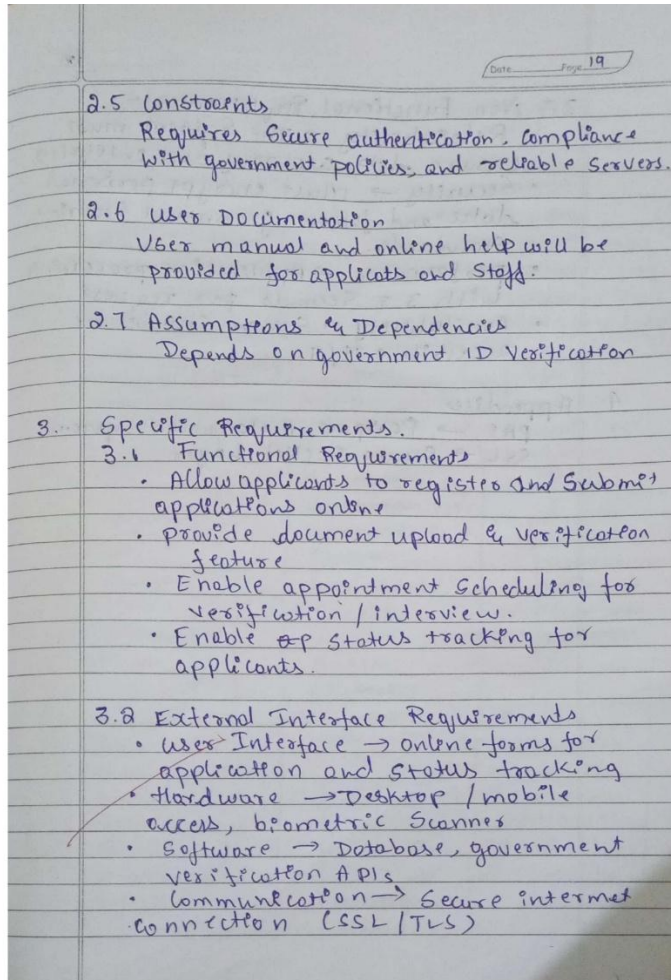
e. Activity Diagram



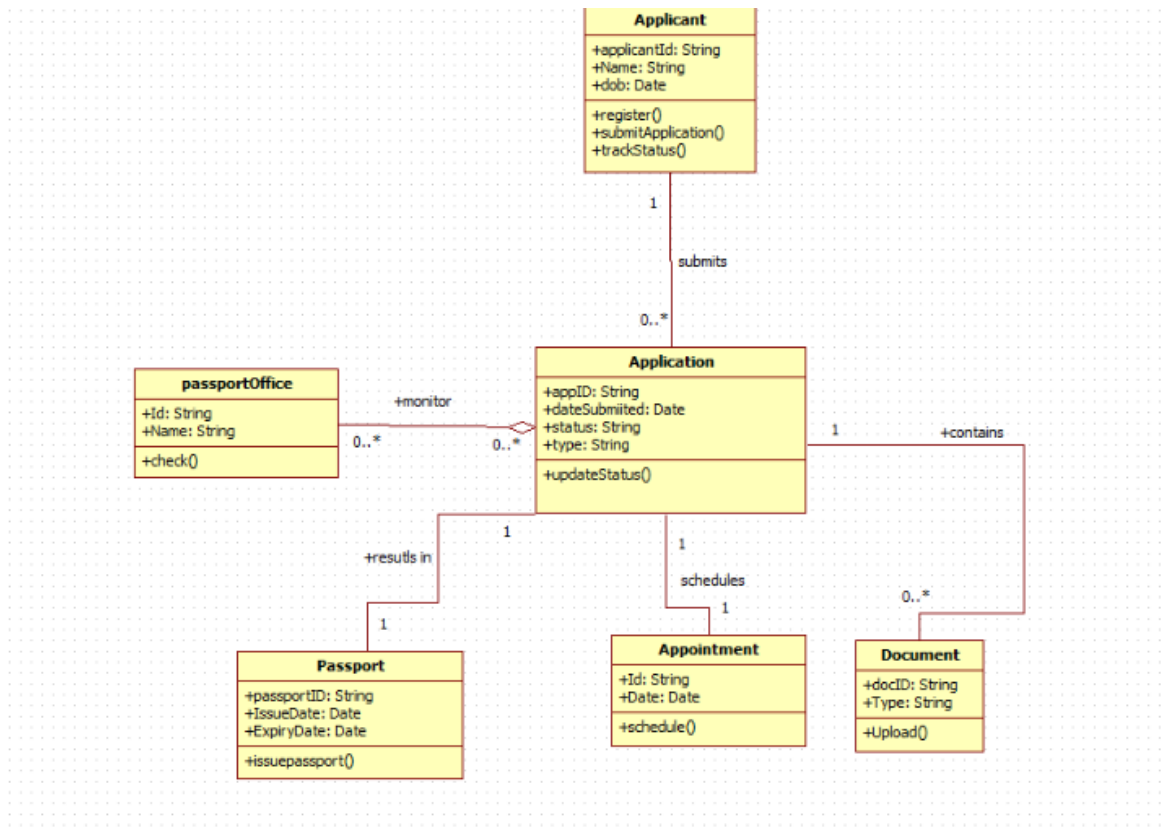
5. Passport Automation System

5.1 SRS Document

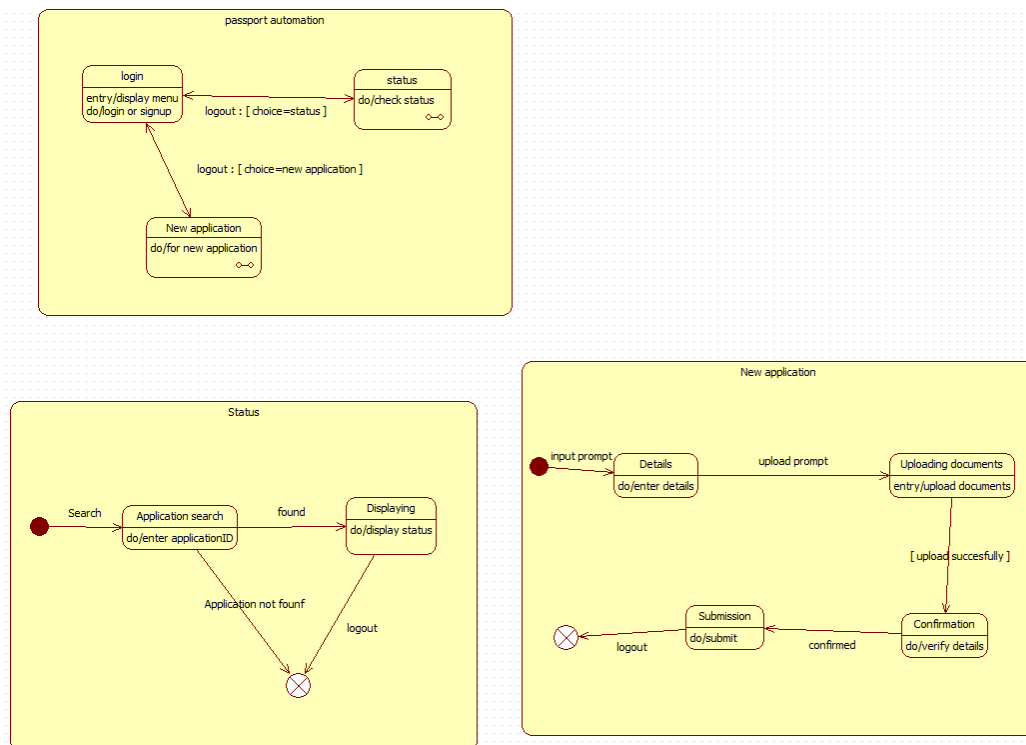




5.2 Class Model

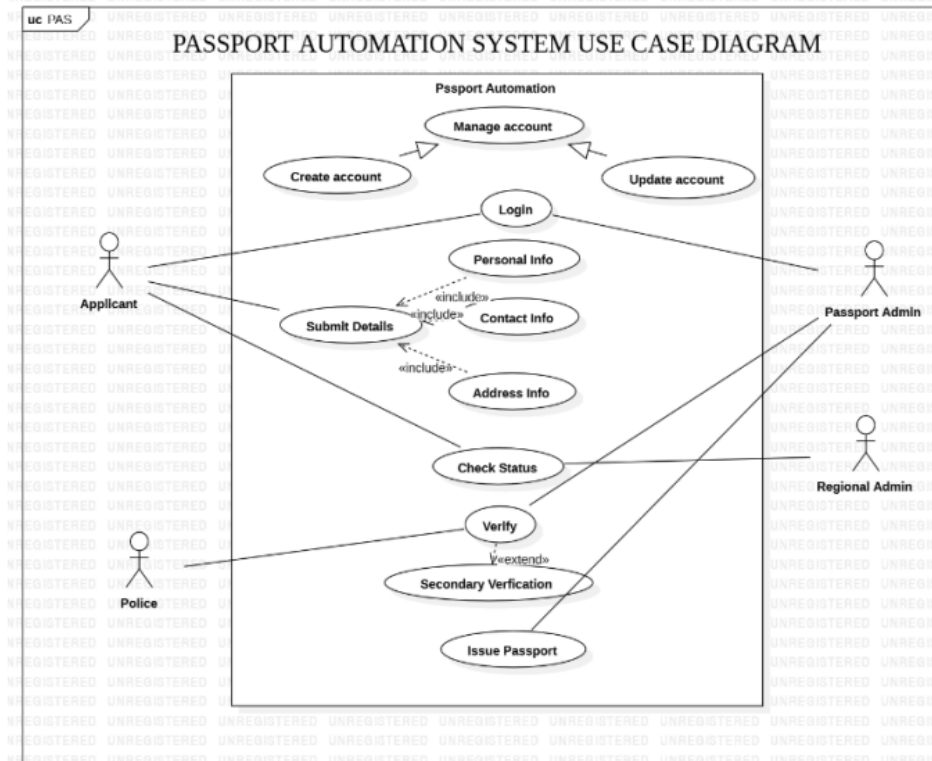


5.3 State Model

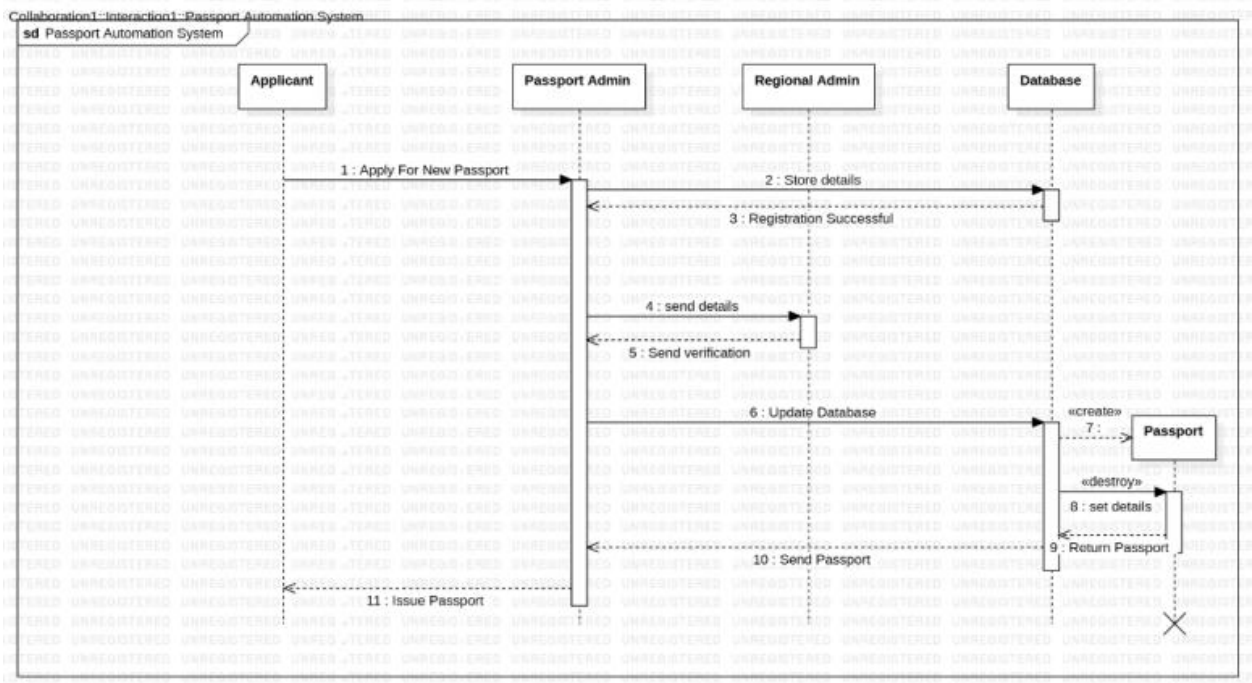


5.4 Interaction Models

a. Use Case Model



a. Sequence Model



b. Activity model

